

EEFE 530 — Problem Set 3

Matching & Propensity Score Methods - Answer Key

EEFE 530: Applied Microeconometrics

Invalid Date

Overview

This problem set uses the classic **LaLonde job training** data to practice matching estimators.

You have two datasets:

- `data/lalonde_nsw.csv`: treated + *experimental* controls (NSW)
- `data/lalonde_psid.csv`: treated + *observational* controls (PSID)

Outcome: `re78` (earnings in 1978).

Treatment: `treat` (1 = participated in training).

Pre-treatment covariates: `age`, `education`, `black`, `hispanic`, `married`, `nodegree`, `re74`, `re75`.

Submission

Push to GitHub Classroom:

- `ps3_<LAST_NAME>.qmd`
- rendered output: `submission_<LAST_NAME>.html` (or `submission_<LAST_NAME>.pdf`)
- any helper files you used (figures, scripts)

Recommended repo structure:

```
ps3_<LASTNAME>.qmd
submission_<LASTNAME>.html
data/
  lalonde_nsw.csv
  lalonde_psid.csv
```

Write your code inline as shown below

```
# Example code
treat.test <- c(1,2,3)
control.test <- c(0,1,2)

att.test <- mean(treat.test) - mean(control.test)
```

1 Load and inspect the data

1.1 Read the CSVs

1.2 Descriptive checks

1. Create a dataset with treated observations and psid controls - the nonexperimental sample.
2. For each dataset, report means of the covariates by treatment status in a well-formatted table.
3. Briefly describe differences in sample sizes, covariates, and outcomes in the two samples. Write this as you would in the data section of an economics paper.

2 Benchmark: experimental ATT in NSW

Treat NSW as an experiment (random assignment inside the NSW dataset).

2.1 Compute the experimental difference in means

Compute:

$$\widehat{ATT}^{exp} = \bar{Y}_1 - \bar{Y}_0$$

using the NSW data only. Use a t-test to test the statistical significance.

Briefly interpret the result.

2.2 Compare to a regression estimate

Create a well-formatted regression tables with three regressions:

1. Simple regression with no controls.
2. Regression with demographics but no pre-treatment outcomes.
3. Regression with demographics and pre-treatment outcomes.

Compare the results in words to the difference in means.

3 Observational comparison

Now use the PSID dataset, which contains treated units plus *observational* controls.

3.1 Naive difference in means

Compute the difference in means in PSID and compare it to the experimental NSW benchmark from Section 2.

3.2 Why does this differ?

In 3–5 sentences, explain why the naive PSID estimate differs from the NSW benchmark.

3.3 Regression

Create a well-formatted regression tables with three regressions:

1. Simple regression with no controls.
2. Regression with demographics but no pre-treatment outcomes.
3. Regression with demographics and pre-treatment outcomes.

Compare the results in words to the difference in means.

3.4 Matching

Create a well-formatted table with the following matching estimates. Estimate the ATT using the `Match` function with the `Matching` package. Set the `BiasAdjust` to `FALSE`.

1. Exact matching on all variables other than pre-treatment outcomes.
2. Exact matching on all variables and quartiles of pre-treatment outcomes and age.
3. Nearest neighbor matching.
4. Nearest neighbor matching with a 0.25 caliper.
5. Matching on 2 and 3 neighbors with 0.25 caliper.

3.5 Matching with Bias Adjustment

Create a well-formatted table with the following matching estimates. Estimate the ATT using the `Match` function with the `Matching` package. Set the `BiasAdjust` to `TRUE`.

1. Exact matching on all variables other than pre-treatment outcomes.
2. Exact matching on all variables and quartiles of pre-treatment outcomes.
3. Nearest neighbor matching.
4. Nearest neighbor matching with a 0.25 caliper.
5. Matching on 2 and 3 neighbors with 0.25 caliper.

Compare the results to 3.4 and interpret what bias-correction does based on [Abadie & Imbens \(2002\)](#).

4 Propensity scores

4.1 Estimate a propensity score model

Estimate a logit model and save the fitted values as the propensity score. Use any variables and interactions that you see fit. Interpret the logit model as it related to casual identification.

4.2 Propensity score diagnostics

1. Plot histograms (or densities) of $\hat{p}(X)$ by treatment status.
2. Comment on overlap.

4.3 Common support & enforcing overlap

Use two methods to define common support:

1. $[\max(\min(\hat{p}_1), \min(\hat{p}_0)), \min(\max(\hat{p}_1), \max(\hat{p}_0))]$ where subscripts indicate treated vs control groups.
2. An additional trimming rule that drops observations with $\hat{p}(X) < 0.1$ or $\hat{p}(X) > 0.9$.

Report how many variables are dropped with each rule.

4.4 PS Matching

Create a well formatted table by matching on the propensity score using the following specifications.

1. Nearest neighbor matching with no overlap.
2. Nearest neighbor matching with trimming rule 1.
3. Nearest neighbor matching with trimming rule 2.
4. Nearest neighbor matching with trimming rule 1 and a 0.25 caliper.

4.5 PS Regression

Create a well-formatted table with propensity score regression with the following specifications.

1. Regress treat with no controls (baseline with selection bias).
2. Add the propensity score as a control with no overlap enforced.
3. Regress with propensity score enforcing overlap rule 1.
4. Regress with propensity score enforcing overlap rule 1 with additional covariates as controls.

5 Extensions

5.1 Genetic Matching

Test out genetic matching in the `Match` package. Compare the ATT using genetic matching and Mahalanobis matching with both 1 and 2 matches. Present results in a well-formatted table.

5.2 Pre-processing

Use `MatchIt` to estimate generate weights that you can use to pre-process the data. Generate weights using both Mahalanobis distance and genetic matching. Evaluate balance between the original dataset and both pre-processed (weighted) datasets.

5.3 Weighting by the propensity score

6 Interpretation

In 4–8 sentences:

- Summarize how well you think the various methods worked to identify causal effects.
- What factors are important to assess when considering using the CIA?
- What are key challenges when using matching?
- How did this assignment affect your intention to use these methods moving forward?